

Groundwater Prolongs but Weakens Meteorological Drought: Evidence from Two Continents

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ABSTRACT: Groundwater drought poses a significant threat, with wide-ranging impacts on society, the environment, and the economy. Despite this, our understanding of groundwater drought has historically been restricted by poor availability of in situ groundwater data. While a common alternative approach has been to use modeling or remote sensing techniques, these methods have significant associated uncertainties. Recent advancements in data sharing mean that it is now possible to address this knowledge gap. Using an extensive dataset of in situ measurements from more than 3200 wells, as well as precipitation data from ERA5-Land, this study investigates groundwater drought at continental scales across North America and Europe. Based on a drought pairing approach, the propagation of meteorological drought to groundwater is assessed using two standardized drought indices. Results indicate that droughts became longer but less intense as they propagated to groundwater. In addition, groundwater droughts showed much greater spatial variability compared to meteorological droughts. Increased drought lengthening was generally observed at sites where the water table was deeper, while greater drought attenuation was typically found at sites where meteorological droughts were more intense. This provides indications of where groundwater is able to more/less effectively buffer meteorological droughts, with important implications for water resource management in these areas.

SIGNIFICANCE STATEMENT: Groundwater drought is a significant threat to human societies. However, our understanding of this phenomenon has historically been restricted by poor availability of direct groundwater measurements. While alternative methods of analysis have been used, these have significant limitations. This study addresses this lack of understanding using a new dataset of direct groundwater measurements from over 3200 wells in North America and Europe. It is found that there were differences in how precipitation deficits were translated to groundwater across both continents. This was linked to the depth of groundwater and differences in the intensity of meteorological droughts at the sites. This has important implications for how water resources are managed in these areas.

KEYWORDS: Europe; North America; Drought; Hydrology; Hydrometeorology; Water resources

1. Introduction

Groundwater is an essential natural resource. It is the primary freshwater source for billions of people, is estimated to provide over 40% of irrigation water globally (Kuang et al. 2024), and is vital to the health of a wide variety of groundwater-dependent ecosystems (Bloomfield et al. 2019; Koirala et al. 2017). Importantly, groundwater is critical to many of the United Nations Sustainable Development Goals (Guppy et al. 2018). For example, effective groundwater management is strongly linked to goal 2, which focuses on ending hunger, improving food security, and promoting sustainable agriculture, as well as goal 6, which seeks to ensure universal access to clean water and sanitation (Velis et al. 2017).

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Groundwater drought, which is defined as a prolonged period of below-normal groundwater levels (Bloomfield et al. 2019), threatens our reliance on groundwater. The impacts can be severe. This can include restrictions on abstractions for domestic, industrial, and agricultural users, which can have costly economic repercussions (Marchant and Bloomfield 2018). For example, a major groundwater drought in the United Kingdom during 2010–12 led to temporary use bans being imposed on around 20 million people, including the agricultural sector in the spring of 2012 (Brauns et al. 2020). Severe meteorological drought conditions occurred in the lead-up to and during this groundwater drought (Kendon et al. 2013). In the Netherlands, a drought in 2018 is estimated to have caused between EUR 450 and EUR 2,080 million in damages due to impacts on farming, navigation, drinking water supplies, and industry (Schutten et al. 2024), many of which were linked to sharply declining groundwater levels (Brakkee et al. 2022). Additionally, there can also be adverse impacts on energy production, tourism, recreation, and river and wetland ecosystems (Brauns et al. 2020; Marchant and Bloomfield 2018).

Meteorological drought is usually the precursor to groundwater drought. Typically, meteorological drought is defined as an extended period of below-normal precipitation (Wilhite and Glantz 1985), although some definitions incorporate high

evaporative demand (e.g., [Beguería et al. 2014](#)). During a prolonged period of below-average precipitation, which may be coupled with above-average evapotranspiration, there is reduced infiltration. As soil moisture depletes, groundwater recharge decreases, leading to the water table becoming deeper. Depending on the antecedent conditions, aquifer characteristics, and nature of the precipitation deficit, this may lead to groundwater drought. This process is termed “drought propagation.” Groundwater typically has a delayed and smoothed response to changes in meteorological conditions, meaning that groundwater drought will not always develop following a dry and/or hot spell. Where this does occur, this can be with considerable delays ([Van Loon 2015](#)). The exact response of groundwater to changes in climate is a complex, site-specific function of both natural and anthropogenic factors ([Bloomfield and Marchant 2013](#); [Kumar et al. 2016](#); [Marchant and Bloomfield 2018](#); [Wendt et al. 2020](#)).

Four key features typically characterize drought propagation. The first is pooling, meaning that multiple meteorological droughts may combine into one prolonged groundwater drought. Attenuation also occurs, whereby the maximum anomaly in the meteorological conditions is smoothed as it moves to groundwater. The third feature is lagging, meaning that there is a delay between the onset of meteorological drought and the resulting groundwater drought. Finally, there is lengthening, with droughts becoming longer ([Van Loon 2015](#)). Having a thorough understanding of these processes, including their influencing factors, is essential for effective water resource management ([Van Lanen et al. 2013](#)) and drought forecasting ([Zhou et al. 2024](#)). This is particularly crucial given that drought frequency is projected to increase in many regions due to climate change ([IPCC 2023a,b](#)).

Groundwater drought has been studied much less extensively than meteorological and streamflow drought. [Christelis et al. \(2025\)](#) discussed a few possible reasons. First, aquifer systems are complex, and the subsurface is highly heterogeneous, making it challenging to produce generalized models of groundwater drought. Second, groundwater responds slowly to changes in climate, meaning that impacts of groundwater drought may take longer to appear compared to other drought types. Additionally, these impacts may not be so directly visible. Finally, this may be due to the historic poor availability of groundwater observations.

Previous studies using in situ data to investigate groundwater drought have generally been limited in spatial extent. For instance, [Bloomfield and Marchant \(2013\)](#) used data from 14 sites across the United Kingdom to find high variability in maximum groundwater drought durations. This was linked to autocorrelation in the recharge signal and intrinsic aquifer properties relating to saturated flow and storage. In contrast, a more extensive analysis was presented by [Kumar et al. \(2016\)](#), where data from 2040 Dutch and German wells were used, showing that the standardized precipitation index (SPI) was not a suitable groundwater drought index. Although the number of wells in this analysis was greater, the spatial extent was again limited. This does not enable comparisons across a diverse range of sites ([Lema et al. 2025](#)).

While some studies have presented larger-scale analyses using in situ data, these have key limitations. For example, [Brunner and Chartier-Rescan \(2024\)](#) used data from central Europe,

showing that 10% of precipitation droughts propagated to groundwater. Although this provided a continental-scale analysis, only 70 sites were considered. Similarly, [Brauns et al. \(2020\)](#) presented a European-scale analysis, demonstrating that hydrogeological properties influence drought propagation. Yet, this was again limited, assessing data from only 16 wells. Although other large-scale studies have considered data from a wider selection of sites, these have limitations in their coverage. For instance, [Schreiner-McGraw and Ajami \(2021\)](#) used data from 266 wells across the United States, showing that shallow aquifers took, on average, 3 years to recover from multiyear droughts. Most sites were in the northeast, with very poor coverage in western regions, making this analysis heavily biased toward these areas.

Due to poor availability of in situ groundwater data, hydrological models have provided an alternative method of analysis ([Li and Rodell 2015](#)). For instance, using a model of the Pang catchment in the United Kingdom, [Peters et al. \(2006\)](#) found spatial differences in groundwater drought characteristics, with short droughts being more severe near the streams. Similarly, applying a transient groundwater model across Germany, [Hellwig et al. \(2020\)](#) showed that upland regions experienced shorter but more frequent droughts than lowland areas. However, model representation of groundwater is imperfect ([Condon et al. 2021](#); [Gleeson et al. 2021](#)). For example, [Van Loon et al. \(2012\)](#) investigated the effectiveness of 10 large-scale models in simulating drought propagation. Key features were correctly simulated, with increases in drought frequency and decreases in drought duration following propagation through the hydrological system. Yet, differences between fast- and slow-responding systems were not adequately represented, indicating that there can be substantial uncertainties associated with using models to assess groundwater drought.

Another alternative approach has been to use data from the Gravity Recovery and Climate Experiment (GRACE). For example, [Han et al. \(2019\)](#) used GRACE data to find that there had been an 8-month propagation time from meteorological drought to groundwater in the Pearl River basin during 2002–15. While such an approach enables analysis over large areas, there are limitations. The spatial resolution of GRACE data is poor ([Marchant and Bloomfield 2018](#)). Furthermore, to assess changes in groundwater storage, terrestrial water storage (TWS) anomalies from GRACE must first be separated into their separate components ([Liu et al. 2023](#)). This is because TWS is comprised of contributions not only from groundwater but also from soil moisture, surface water, ice, snow, and biomass. This requires the use of either in situ data or land surface models ([Gleeson et al. 2021](#); [Rodell et al. 2009](#)), therefore introducing additional uncertainty into the analysis.

There have been significant advancements in the availability of in situ groundwater data in recent years. This is largely due to the work of the International Groundwater Resources Assessment Centre (IGRAC), which has promoted data sharing between countries and supported the development of new monitoring networks ([IGRAC 2025a](#)). IGRAC have compiled data from tens of thousands of sites into one freely available dataset. Compared to previous studies, which have predominantly focused on analysis using in situ data over smaller regions, this dataset presents an excellent opportunity

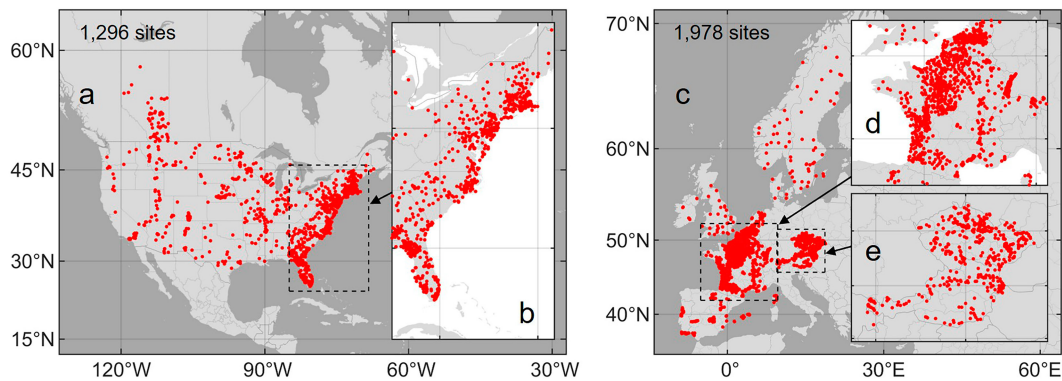


FIG. 1. Wells in (a) North America, with (b) an inset showing detail along the eastern coast, and (c) Europe, with insets showing detail in (d) France/Belgium and (e) Austria/Czechia. Dashed lines indicate the regions enlarged in the insets.

to assess groundwater drought across sites reflecting a wide range of climatic and hydrogeological settings. While large-scale analysis is also possible using GRACE data, IGRAC's dataset provides direct observations that do not require separation from other TWS components.

Therefore, using this dataset, this study aimed to address this knowledge gap by investigating the variability of groundwater drought at continental scales using in situ data. In the following, the spatial variability in drought characteristics is first assessed. Then, the propagation of meteorological drought to groundwater is analyzed using a drought pairing approach. Finally, influencing factors of drought propagation are investigated.

2. Data and methods

a. Groundwater and precipitation data

Publicly available groundwater data were obtained from IGRAC's Global Groundwater Monitoring Network (GGMN) for 51 903 wells across North America and Europe (IGRAC 2025b). Measurements were provided as groundwater table depths or groundwater level elevations. In the latter case, these were converted to water-table depth using elevation data from version 1.03 of the Multi-Error-Removed Improved-Terrain (MERIT) DEM (Yamazaki et al. 2017), downloaded using the Google Earth Engine (Gorelick et al. 2017). Because the frequency of observations varied, the format of each time series was standardized using monthly mean values. Sites with negative overall mean water-table depths were discarded; this indicates that water levels were above the ground surface and were therefore not representative of a groundwater table (Baird and Low 2022). These were mostly sites where water-table elevations had been converted.

Time series from the remaining sites covered a range of periods, and many had extended intervals with no measurements. These were first filtered to only include sites with observations which both (i) started before or during January 2000 and (ii) ended during or after December 2019. Consistency in the length of each time series was important as this can influence drought characterization (Wu et al. 2005); therefore, only measurements recorded between these dates were considered. Although it is typically recommended that 30 years of data

should be used for estimating drought indices (Kumar et al. 2016; WMO 2012), 20 years was selected as a compromise with the number of wells that would be suitable for analysis. This approach has been applied in other studies using in situ data to investigate groundwater drought (e.g., Kumar et al. 2016).

Certain requirements for time series completeness were also specified. Following the approach of Wendt et al. (2020), short periods of consecutive months of missing data (less than 6 months) were permissible. Over the 20 years, however, less than 20% of values were permitted to be missing overall. These requirements were similarly a compromise to ensure a reasonable sample size of wells. Based on an assessment of the suitability of various gap-filling methods (see the online supplemental material), gaps less than 6 months were filled using modified Akima interpolation (Akima 1970).

For assessing meteorological drought, monthly precipitation data for each site were obtained from ERA5-Land. This is a global reanalysis dataset with a spatial resolution of 0.1° (Muñoz-Sabater et al. 2021). It was not possible to obtain data for a small number of coastal sites, and so these were removed from further analysis. Following this filtering, 3274 wells were suitable, with the locations shown in Fig. 1.

b. Drought indices

Standardized drought indices were used to identify meteorological and groundwater droughts. This approach involves comparing current conditions to long-term averages using standardized scales (Van Loon 2015). This was preferred over fixed or variable threshold level approaches, where droughts are instead identified using thresholds typically based on percentiles (e.g., Peters et al. 2006; Van Loon and Laaha 2015), due to enabling more straightforward comparisons of droughts across varying regions and climates (Van Loon 2015). Meteorological drought was identified using the SPI. Originally proposed by McKee et al. (1993), the SPI expresses deviations in the wetness/dryness of a site relative to its long-term mean conditions. The standardized groundwater index (SGI), which was later developed by Bloomfield and Marchant (2013), is analogous to the SPI but considers groundwater data. Both have been extensively used for drought identification and analysis (Dubois and Larocque 2024; Mishra and Singh 2010).

To calculate the SPI for a site, monthly precipitation is first accumulated over a predetermined number of months. In this study, the SPI was calculated using a 3-month accumulation period. This is commonly used to characterize short- and medium-term meteorological droughts (Cammalleri et al. 2023). Subsequently, the resulting values are split up by calendar month. For a given accumulated precipitation in each calendar month, the cumulative probability of observing that particular value is determined. Finally, the inverse normal cumulative distribution function is applied to these probabilities to determine the SPI (Bloomfield and Marchant 2013; McKee et al. 1993).

Calculation of the SGI is very similar with a few differences. Because groundwater table depth is a continuous variable, it is not accumulated. Also, a normal scores transform is used to determine the cumulative probability of a certain value in each calendar month. This is instead of estimating the probability through distribution fitting, which is the approach commonly used when calculating the SPI. In the first step of calculating the SGI, the groundwater data are split up by calendar month. Next, for a time series spanning n years, the inverse normal cumulative distribution function is applied to n equally spaced probabilities between $1/(2n)$ and $1 - 1/(2n)$. The resulting values are the SGI. Finally, for measurements taken during a given calendar month over all years of the time series, higher values of the SGI are matched to lower water-table depths; i.e., the SGI is higher when the water table is shallower (Bloomfield and Marchant 2013). Following the approach of Bloomfield and Marchant (2013), both the SPI and SGI were calculated using a normal scores transform to ensure consistency between the two indices.

Droughts are identified when the SPI/SGI falls below a threshold. Commonly, this is set as 0 (e.g., Bloomfield and Marchant 2013; Guo et al. 2021) or -1 (e.g., Fuentes et al. 2022; Marchant and Bloomfield 2018; Xiong et al. 2025). While a threshold of 0 defines drought as any period of below-normal water availability, a threshold of -1 means that a more substantial departure from normal conditions is necessary for drought to occur. This significantly impacts the time spent in drought, with a threshold of 0 leading to around three times as many drought months being detected for any given site. However, the additional droughts identified using a threshold of 0 may not meaningfully link to drought impacts, for example, on agriculture, water supply, energy, and industry (Bachmair et al. 2016). Therefore, a value of -1 was adopted to maximize the chances of identified droughts corresponding to observable impacts. Additionally, consecutive droughts with less than 3 months between them were pooled (Xiong et al. 2025). This minimizes the disruption caused by minor droughts to the statistical analysis of drought propagation (Guo et al. 2020; Xiong et al. 2025).

Figure 2 shows how drought characteristics can be determined; this approach is identical if instead considering the SPI. Drought duration is the number of consecutive months during which the SPI/SGI is below the drought threshold. Drought severity is the sum of SPI/SGI values below the threshold (the area shown in dark green in Fig. 2). Finally, intensity is the ratio between severity and duration (Xiong et al. 2025).

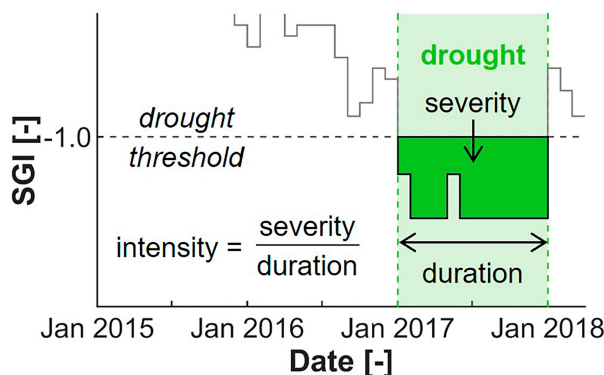


FIG. 2. An example time series showing how drought characteristics are extracted from the SGI. This approach is identical when considering the SPI instead.

c. Drought pairing

Drought propagation was investigated by pairing groundwater droughts with their corresponding meteorological droughts. This is an adapted version of the approach presented by Xiong et al. (2025) to assess the propagation of meteorological drought to streamflow. For a given groundwater drought, the paired meteorological drought is the one that started between the onset of that groundwater drought and the onset of the previous groundwater drought. In the cases where multiple meteorological droughts met this criterion, the one with the latest onset was selected (Xiong et al. 2025). This assumes that groundwater droughts are triggered by only the latest occurring meteorological drought. Figure 3 illustrates this approach. In this figure, two drought pairs are indicated in yellow and red, with the nonpaired meteorological drought shown in gray. The merging of consecutive droughts with less than 3 months between is also illustrated. The vast majority (96.8%) of sites had at least one drought pair, with most (57.4%) having five or more.

d. Drought propagation ratios

Propagation ratios were used to investigate the impacts of drought propagation on the drought characteristics. These are defined as the ratio of the groundwater drought characteristic to the paired meteorological drought characteristic and can be calculated using Eqs. (1)–(3) (Xiong et al. 2025):

$$R_D = \frac{D_G}{D_M}, \quad (1)$$

$$R_I = \frac{I_G}{I_M}, \quad (2)$$

$$R_S = \frac{S_G}{S_M}, \quad (3)$$

where D_G , I_G , and S_G are the duration (months), intensity (—), and severity (—) of the groundwater drought, respectively; D_M , I_M , and S_M are the duration (months), intensity (—), and severity (—) of the paired meteorological drought, respectively; and R_D , R_I , and R_S are the propagation ratios of

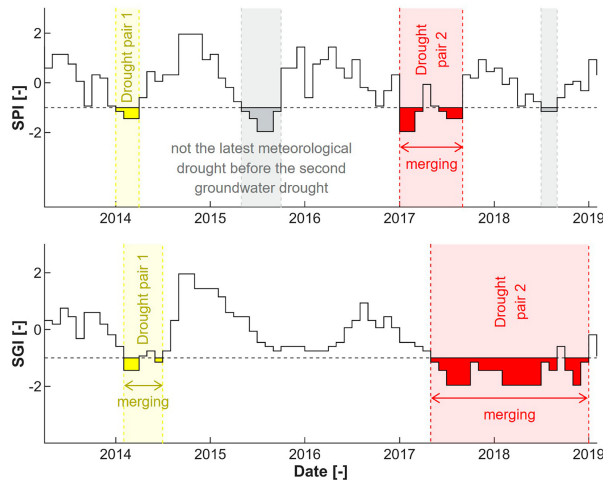


FIG. 3. Example SPI and SGI time series from a given site showing the pairing of meteorological droughts to groundwater droughts, adapted from Xiong et al. (2025).

duration, intensity, and severity (—), respectively. Following the approach of Xiong et al. (2025), a symmetrical transformation is performed on these values using Eqs. (4)–(6) to produce values ranging between -1 and 1 :

$$R_D^* = \begin{cases} \frac{R_D - 1}{R_D}, & R_D \geq 1 \\ R_D - 1, & R_D < 1 \end{cases}, \quad (4)$$

$$R_I^* = \begin{cases} \frac{R_I - 1}{R_I}, & R_I \geq 1 \\ R_I - 1, & R_I < 1 \end{cases}, \quad (5)$$

$$R_S^* = \begin{cases} \frac{R_S - 1}{R_S}, & R_S \geq 1 \\ R_S - 1, & R_S < 1 \end{cases}, \quad (6)$$

where R_D^* , R_I^* , and R_S^* are the adjusted drought propagation ratios (—). Positive values of R_D^* , R_I^* , and R_S^* indicate increases in the drought characteristic following propagation, negative values signify decreases, and ratios of 0 show no change. For example, a positive duration propagation ratio would imply that the groundwater drought was longer than its corresponding meteorological drought, a negative ratio would show that the groundwater drought was shorter, and a value of zero would indicate equal lengths.

3. Results

a. Mean meteorological drought characteristics

Figure 4 shows the variation in mean meteorological drought durations, intensities, and severities across North America and Europe. Some spatial variability can be observed. For example, there are regions where meteorological droughts tended to be

longer (Figs. 4a,b), more severe (Figs. 4c,d), and less intense (Figs. 4e,f) compared to surrounding areas, such as in northern Mexico, southwestern Spain, and northern Italy. However, the variability across sites was not substantial, remaining within the same order of magnitude for all three drought characteristics.

b. Mean groundwater drought characteristics

Figure 5 shows the mean groundwater drought durations, intensities, and severities at each well. Each shows high spatial variability. This is particularly clear for the mean drought durations and severities, both of which varied by over an order of magnitude across the sites. This spatial variability can be seen, for example, along the eastern coast of North America (Figs. 5b,l). In the southeast of this region, there is a cluster of wells with much higher drought durations and severities compared to those at the surrounding sites. Similarly, there is a large group of wells in northern France and Belgium (Figs. 5d,n) where droughts were longer and more severe than those in neighboring areas. While spatial variability was also observed in the mean groundwater drought intensities, this was much less pronounced, varying across a much narrower range of values. In Czechia, for example, droughts were slightly more intense in the northeast than in the southern and southwestern areas (Fig. 5j).

Visual inspection of Figs. 4 and 5 indicates that the high spatial variability in groundwater drought characteristics could not be explained by that in the corresponding meteorological drought characteristics, and there was no consistent relationship between the two. This implies that a site having longer, more intense, or more severe meteorological droughts did not always correspond to that site having longer, more intense, or more severe groundwater droughts. Along the eastern coast of North America, for example, the high spatial variability in groundwater drought durations and severities is not reflected in the corresponding meteorological drought characteristics. Similarly, in northern France and Belgium, where groundwater drought durations were higher than those in the surrounding areas, meteorological droughts were not noticeably longer. This lack of consistent relationships is further supported by very weak correlations between the meteorological drought characteristics and the corresponding groundwater drought characteristics in each case (supplemental material).

c. Quantitative analysis of drought propagation ratios

Figure 6 shows the mean propagation ratios of duration, intensity, and severity at each site. These quantify how the drought characteristics changed during propagation to groundwater, with negative values indicating decreases and vice versa. Figures 6a–e show high spatial variability in the mean duration propagation ratio. This is particularly clear along the eastern coast of North America (Fig. 6b). Across this region, there is a pronounced difference in propagation ratios at wells further north, where these tended to be close to 0, and further south, where values were generally much greater. This implies that there was typically little change in duration at these more northern sites; however, groundwater droughts were longer than the corresponding meteorological droughts at the sites further south. In northern France and eastern Austria (Figs. 6d,e),

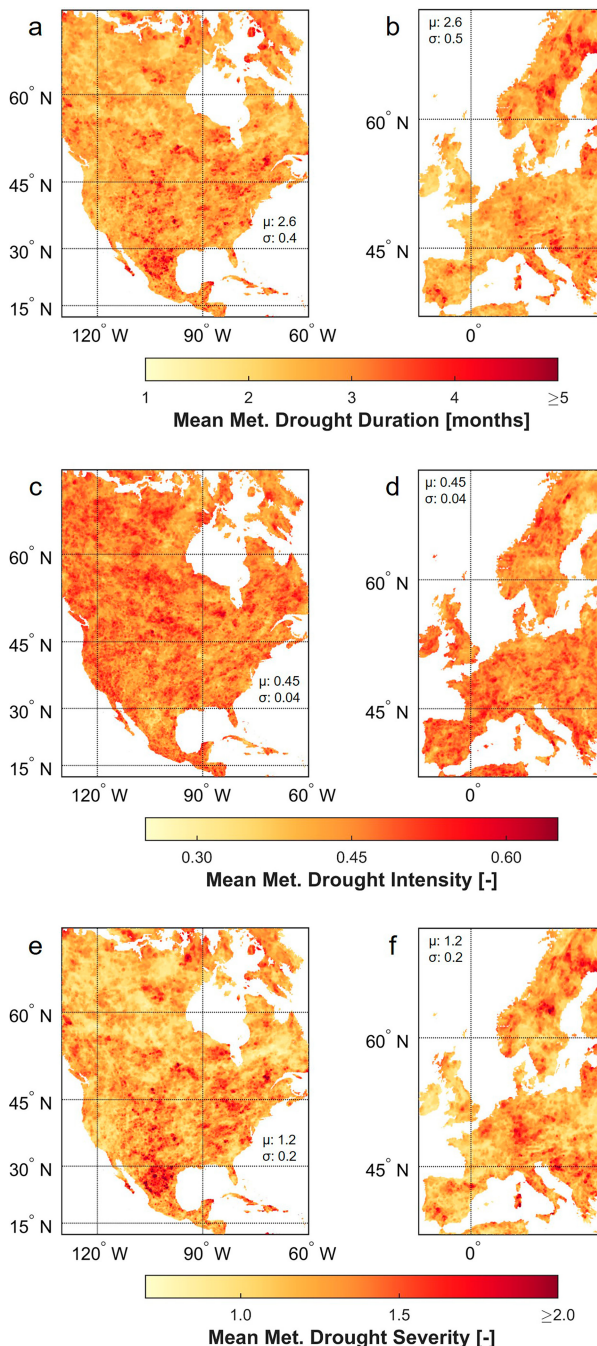


FIG. 4. Mean meteorological drought durations in (a) North America and (b) Europe; mean meteorological drought intensities in (c) North America and (d) Europe; and mean meteorological drought severities in (e) North America and (f) Europe. The mean and standard deviation are indicated in each case by μ and σ , respectively.

there are similarly groups of wells with high propagation ratios, indicating large increases in durations in these areas.

Spatial patterns can also be observed in the mean intensity propagation ratio in Figs. 6f–j, although these are much less pronounced. For instance, there are very small clusters of

wells with high positive propagation ratios along the eastern coast of North America (Fig. 6g) and in northern France (Fig. 6i). This implies that groundwater droughts were much more intense than the corresponding meteorological droughts at these sites. Conversely, there are groups of wells with large negative intensity propagation ratios in Belgium and southeastern Czechia (Fig. 6j), indicating a substantial reduction in intensity.

Significant spatial variability was similarly observed in the mean severity propagation ratio in Figs. 6k–o. Along the eastern coast of North America (Fig. 6l), there is again a distinct difference in propagation ratios at sites further north, where these tended to be close to zero, and further south, where these were generally high and positive. For the more northern sites, this suggests minimal changes in severity following propagation. Conversely, groundwater droughts tended to be much more severe than the corresponding meteorological droughts at the sites further south. Similarly, there are groups of wells with high positive propagation ratios in northern France and low negative propagation ratios in Belgium (Fig. 6n), implying large increases and decreases in severity, respectively.

To further quantify their distributions, Fig. 7 shows the empirical cumulative distribution functions of the duration, intensity, and severity propagation ratios in North America and Europe. The distribution of values across North American sites falls below that of the European sites for all three propagation ratios, with this being most pronounced for duration and severity. This indicates that all three tended to be lower at sites in Europe.

Figure 7 also shows that the majority of sites had a positive mean duration propagation ratio, with most of the distribution being between 0 and 1 in both continents (70% and 68% for North America and Europe, respectively). This implies that drought durations lengthened at most sites. Conversely, sites mainly had negative mean intensity propagation ratios, with 65% and 71% of the distribution falling between -1 and 0 for North America and Europe, respectively, suggesting that intensity typically reduced. The mean severity propagation ratio was positive at only a slightly larger proportion of sites in both continents (57% and 51% for North America and Europe, respectively), implying that propagation had a very inconsistent impact on drought severity.

Relationships between the propagation ratios and a range of factors, including the geology, topography, and climate characteristics, were assessed. Links were found with the mean meteorological drought characteristics and the mean water-table depth; these are presented in the following. For the other factors, this analysis is provided in the supplemental material.

d. Relationships with mean meteorological drought characteristics

To explore the effects of meteorological drought characteristics on drought propagation, Fig. 8 shows the relationships between the mean meteorological drought duration, intensity, and severity and the corresponding propagation ratios. Figure 8a shows a negative relationship between the mean meteorological drought duration and the mean duration propagation ratio. This was a weak relationship, with substantial scatter and a weak correlation coefficient. Additionally, as demonstrated by

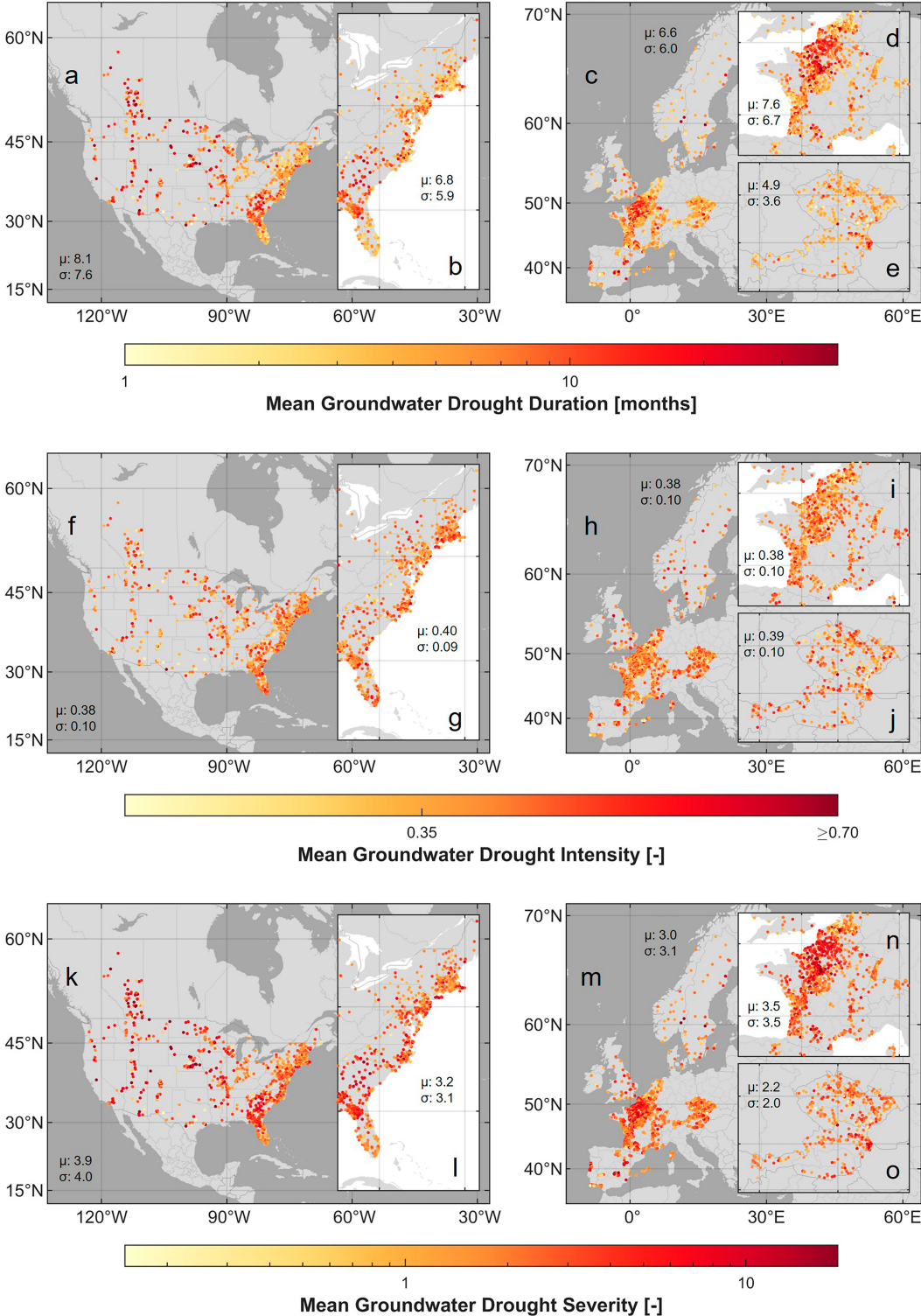


FIG. 5. Mean groundwater drought durations in (a) North America and (c) Europe; mean groundwater drought intensities in (f) North America and (h) Europe; and mean groundwater drought severities in (k) North America and (m) Europe. Insets show detail along the east coast of (b),(g),(l) North America, (d),(i),(n) France/Belgium, and (e),(j),(o) Czechia/Austria. The mean and standard deviation are indicated in each case by μ and σ , respectively.

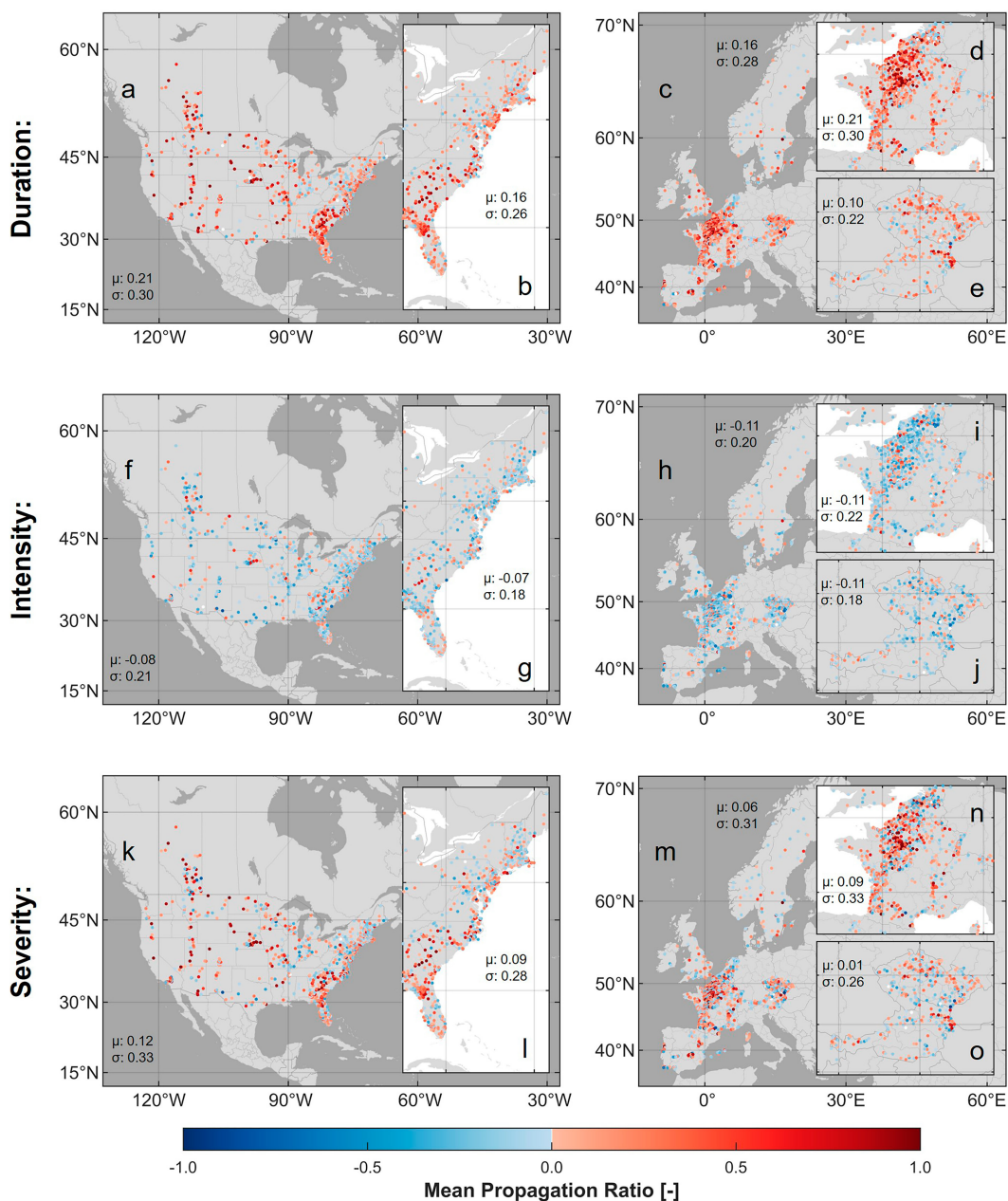


FIG. 6. Mean duration propagation ratios in (a) North America and (c) Europe; mean intensity propagation ratios in (f) North America and (h) Europe; and mean severity propagation ratios in (k) North America and (m) Europe. Insets show detail along the east coast of (b),(g),(l) North America, (d),(i),(n) France/Belgium, and (e),(j),(o) Czechia/Austria. The mean and standard deviation are indicated in each case by μ and σ , respectively.

the median values being positive across all bins, most sites experienced increases in drought duration following propagation, regardless of the duration of the meteorological droughts at the site. However, the negative relationship means that the propagation ratio tended to decrease with increasing mean meteorological drought duration, becoming closer to zero at sites with very long meteorological droughts. This implies that there was typically less drought lengthening at sites experiencing longer meteorological droughts.

Figure 8b shows a much stronger negative relationship between the mean meteorological drought intensity and the mean intensity propagation ratio. The proportions of sites experiencing increases/decreases in intensity varied considerably across the mean meteorological drought intensities considered. Of the sites with the least intense meteorological droughts (below the 20th percentile), most (70%) had a positive propagation ratio, indicating intensity amplification. However, the propagation ratio tended to become increasingly negative with increasing

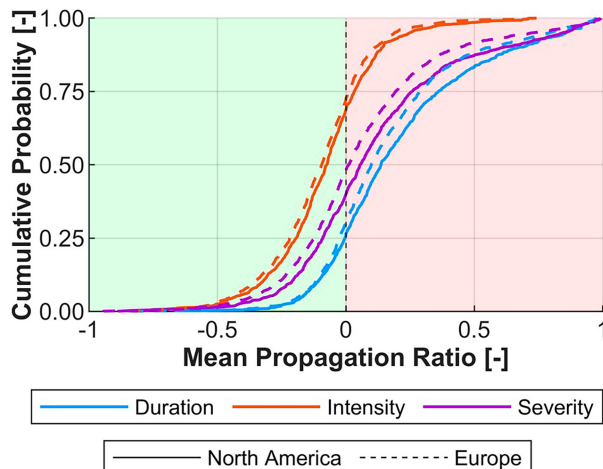


FIG. 7. Empirical cumulative distributions of the mean duration, intensity, and severity propagation ratios for sites in North America and Europe. Positive and negative values are delineated, with the red and green areas indicating increases and decreases in the drought characteristic being considered, respectively.

meteorological drought intensity. Of the sites where meteorological droughts were most intense (above the 80th percentile), almost all (98%) had a negative propagation ratio, signifying a reduction in intensity. This negative relationship means that higher mean meteorological drought intensity was associated with greater decreases in intensity following propagation to groundwater.

A negative relationship was also found between the mean meteorological drought severity and the mean severity propagation ratio in Fig. 8c. This was the weakest relationship, with high scatter and a low correlation coefficient. There was some variation in the proportions of sites experiencing increases/decreases in severity over the range of mean meteorological drought severities. Most (78%) sites with the least severe meteorological droughts (below the 20th percentile) had a positive propagation ratio, indicating that severity tended to increase. At higher mean meteorological drought severities, the proportions of sites with positive and negative propagation ratios were fairly equal, with the median close to zero in all cases, and minimally changed with increasing meteorological drought severity. Therefore, with the exception of the sites with the least severe meteorological droughts, the propagation ratio was not clearly related to meteorological drought severity.

e. Relationships with mean water-table depth

The influence of water-table depth on the propagation ratios was also assessed. Figure 9 shows the mean water-table depth (WTD) at each site. This significantly varied, ranging from shallow (mean WTD < 10 m) to very deep (mean WTD > 500 m). There were some spatial patterns. In North America, the water table tended to be shallower along the east coast yet deeper in southwestern areas. In Europe, areas of high mean WTD can be seen in northern France and the southeast of the United Kingdom, whereas much shallower water tables can be found in Belgium and Czechia.

Relationships between the mean water-table depth and the mean duration, intensity, and severity propagation ratios are shown in Fig. 10. Figure 10a indicates a positive relationship between the mean WTD and the mean duration propagation ratio. This means that sites with higher mean WTDs, and therefore deeper groundwater, tended to experience greater lengthening of drought durations during propagation. In contrast, sites with very shallow groundwater (mean WTD < 5 m) typically had propagation ratios that were closer to zero, implying that meteorological and groundwater drought durations were generally more similar in these areas. However, as indicated by the substantial scatter and low correlation coefficient, this was not a strong relationship. Figure 10a also shows that there was greater variability in propagation ratios in areas with deeper groundwater, with a hugely increasing dispersion of values in each bin with increasing mean WTDs.

In contrast, Fig. 10b shows no clear relationship between the mean WTD and the mean intensity propagation ratio, with high scatter and a very low correlation coefficient. Additionally, the median values of each bin minimally change with increasing mean WTD, and the spread of values in each is fairly consistent. This suggests that changes in intensity following drought propagation were unrelated to the mean WTD.

Figure 10c shows a positive relationship between the mean WTD and the mean severity propagation ratio. This implies that sites with deeper groundwater typically saw greater increases in severity following drought propagation. Conversely, the propagation ratio tended to be closer to zero at sites with very shallow groundwater (mean WTD < 5 m), suggesting that meteorological and groundwater drought severities were similar at these sites. Figure 10c also shows much greater variability in the mean severity propagation ratio with increasing mean WTD, with the range of values in each bin substantially increasing. Once again, this relationship was weak, with a low correlation coefficient and substantial scatter.

4. Discussion

a. Spatial variability of groundwater drought characteristics

Mean groundwater drought durations, intensities, and severities were determined for 3274 sites across North America and Europe. This has provided novel insights into the variability of groundwater drought at continental scales using a new global dataset of in situ groundwater observations. This is in contrast to previous studies, which have typically analyzed groundwater drought using observational data over much smaller regions. This broader analysis has therefore enabled investigations into groundwater drought at sites covering a much wider range of climatic and hydrogeological conditions.

Significant spatial variability was observed in the groundwater drought characteristics in both continents, with this being particularly pronounced for the mean drought durations and severities. The slightly lower spatial variability observed in the mean groundwater drought intensities can likely be attributed to the very high correlation between groundwater drought severity and duration ($\rho = 0.94$, supplemental material). Because

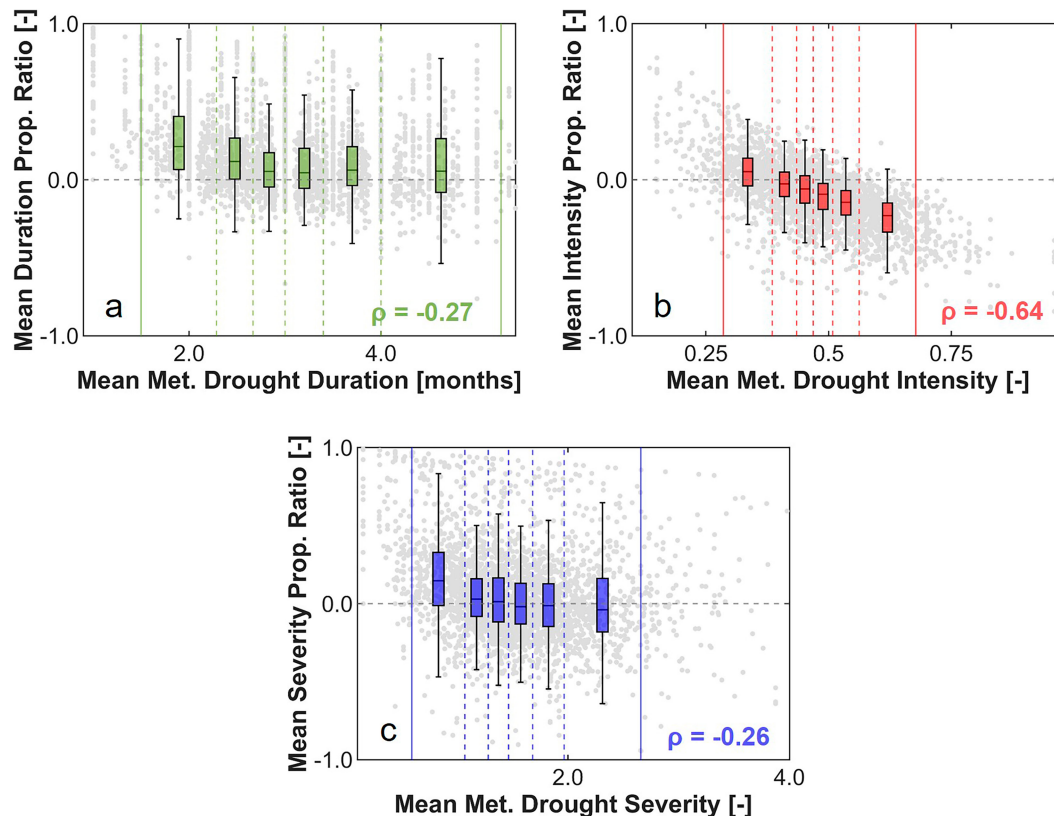


FIG. 8. Relationships between (a) mean meteorological drought duration and the mean duration propagation ratio; (b) mean meteorological drought intensity and the mean intensity propagation ratio; and (c) mean meteorological drought severity and the mean severity propagation ratio. Each gray point represents data from an individual site. The data have been binned between the 5th and 95th percentiles, with equal numbers of sites in each, due to extreme outliers in the mean meteorological drought duration and severity. A small number of sites are not within the limits of these plots. Spearman rank correlation coefficients ρ are indicated in each case; these have been calculated using all data points and, in all cases, were significant at the 5% significance level.

intensity is defined as the ratio between the two, this reduces the occurrence of extreme values, leading to lesser spatial heterogeneity. Overall, the high spatial variability observed in the groundwater drought characteristics is consistent with the findings of many previous smaller-scale studies, for example, in the United States (e.g., Schreiner-McGraw and Ajami 2021; Yin et al. 2022), Canada (e.g., Dubois and Larocque 2024), China (e.g., Huang et al. 2021), Taiwan (e.g., Chu 2018), Austria, Sweden, France (e.g., Brauns et al. 2020), and the United Kingdom (e.g., Bloomfield et al. 2015; Bloomfield and Marchant 2013; Brauns et al. 2020; Peters et al. 2006).

Meteorological drought characteristics were much less spatially variable. This means that the high spatial variability in groundwater drought characteristics could not be fully explained by that observed in the meteorological drought characteristics. This is consistent with the widely accepted understanding that the exact response of groundwater to changes in meteorological conditions is site specific: This can be influenced by not only climatic conditions but also site and aquifer characteristics, as well as anthropogenic activities such as abstraction, thus leading to pronounced spatial heterogeneity (Bloomfield and Marchant 2013;

Kumar et al. 2016; Marchant and Bloomfield 2018; Wendt et al. 2020). Conversely, meteorological drought is often linked to drivers which vary over much larger scales, for example, atmospheric circulation patterns (Kingston et al. 2015; Schubert et al. 2016) and sea surface temperatures (Schubert et al. 2016).

b. Drought propagation

1) COMPARISONS WITH CONCEPTUAL UNDERSTANDING

In general, the four key features that are typically conceptualized as characterizing drought propagation (pooling, attenuation, lagging, and lengthening) have broadly been reflected in this study. Some were accounted for within the drought identification and pairing approach. For example, pooling was considered by merging consecutive meteorological droughts with less than 3 months between them. This approach accounts for the fact that larger droughts are often separated into multiple smaller droughts which may be mutually dependent (Guo et al. 2020). Overall, around a quarter (26.6%) of drought pairs involved a merged meteorological drought, indicating that pooling was fairly common, yet far from universal. Lagging was also

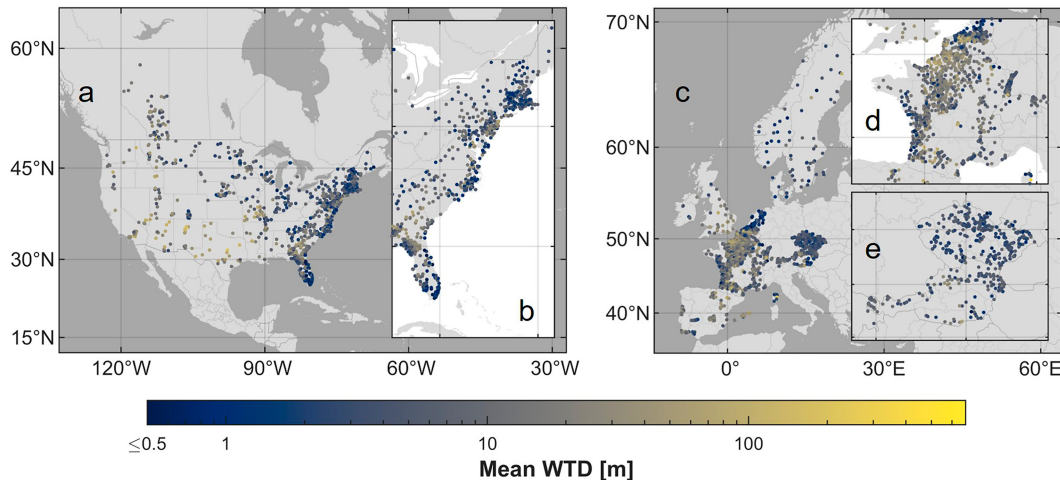


FIG. 9. Mean WTD at sites in (a) North America, with (b) an inset showing detail along the eastern coast, and (c) Europe, with insets showing detail in (d) France/Belgium and (e) Austria/Czechia.

considered in the pairing process, with the first meteorological drought occurring after the onset of the groundwater drought being selected, thereby assuming that it takes at least 1 month for drought propagation to occur. Figure 11 shows the distribution of the mean lag time between the onset of meteorological droughts and the corresponding groundwater droughts. This greatly varied, with a minimum of 1 month, a maximum of 31 months, and a median of 5.8 months. Large variations in propagation times between sites have also been widely reported in many previous studies (e.g., Liu et al. 2023; Nygren et al. 2022; Zhang et al. 2021), further emphasizing the site-specific nature of the response of groundwater to meteorological drought.

The remaining propagation features are evident within the results. First, the mean intensity propagation ratio was negative at most (68.7%) sites. This indicates that there were typically reductions in intensity following propagation, thus showing drought attenuation. Similarly, the mean duration propagation ratio was positive at most (69.3%) sites, implying that drought lengthening typically occurred. Therefore, all four propagation features were broadly represented, indicating that the findings of this study are generally consistent with the conceptual understanding of drought propagation.

The inconsistent changes in severity following drought propagation can likely be explained by the combined effects of drought lengthening and attenuation. This can be understood by examining the relationships between the propagation ratios. There was a very strong positive correlation between the mean severity propagation ratio and the mean duration propagation ratio ($\rho = 0.81$, supplemental material). Similarly, there was a moderately strong correlation between the mean severity propagation ratio and the mean intensity propagation ratio ($\rho = 0.53$). This means that a higher mean severity propagation ratio was associated with both (i) greater drought lengthening, as indicated by a higher mean duration propagation ratio, and (ii) reduced drought attenuation, as implied by a higher intensity propagation ratio. However, the correlation between the mean duration propagation ratio and the mean intensity propagation ratio was

very weak ($\rho = 0.11$). This means that the degree of drought lengthening at any given site was unrelated to the extent to which droughts were attenuated, leading to these inconsistent changes in drought severity.

2) LINKS WITH MEAN METEOROLOGICAL DROUGHT CHARACTERISTICS

There was a very strong negative relationship between the mean intensity propagation ratio and the mean meteorological drought intensity. This implies that greater drought attenuation typically occurred in areas where meteorological droughts were more intense. Conversely, relationships between the mean meteorological drought durations and severities and the corresponding propagation ratios were much weaker, therefore suggesting a lack of influence on drought lengthening.

The greater reductions in drought intensity at sites where meteorological droughts were more intense may be linked to physical mechanisms which affect groundwater's response to ongoing precipitation deficits. During a period of meteorological drought, and in the absence of human interventions, the groundwater system receives less recharge (Van Loon 2015). As the water table deepens, groundwater outflow in the form of baseflow decreases (Eltahir and Yeh 1999; Van Loon 2015). Additionally, water tables can become inaccessible to vegetation, reducing groundwater uptake by plants with insufficient rooting depths (Schreiner-McGraw and Ajami 2021). Both processes may reduce rates of further groundwater level declines, therefore restricting additional increases in groundwater drought intensity. Thus, the effects of these nonlinear processes may prevent groundwater drought intensities from reaching those of the driving meteorological droughts. Given that meteorological droughts are projected to intensify in some regions (IPCC 2023b), having an improved understanding of the potential influence of these nonlinear responses may therefore become increasingly important in assessing future groundwater vulnerability.

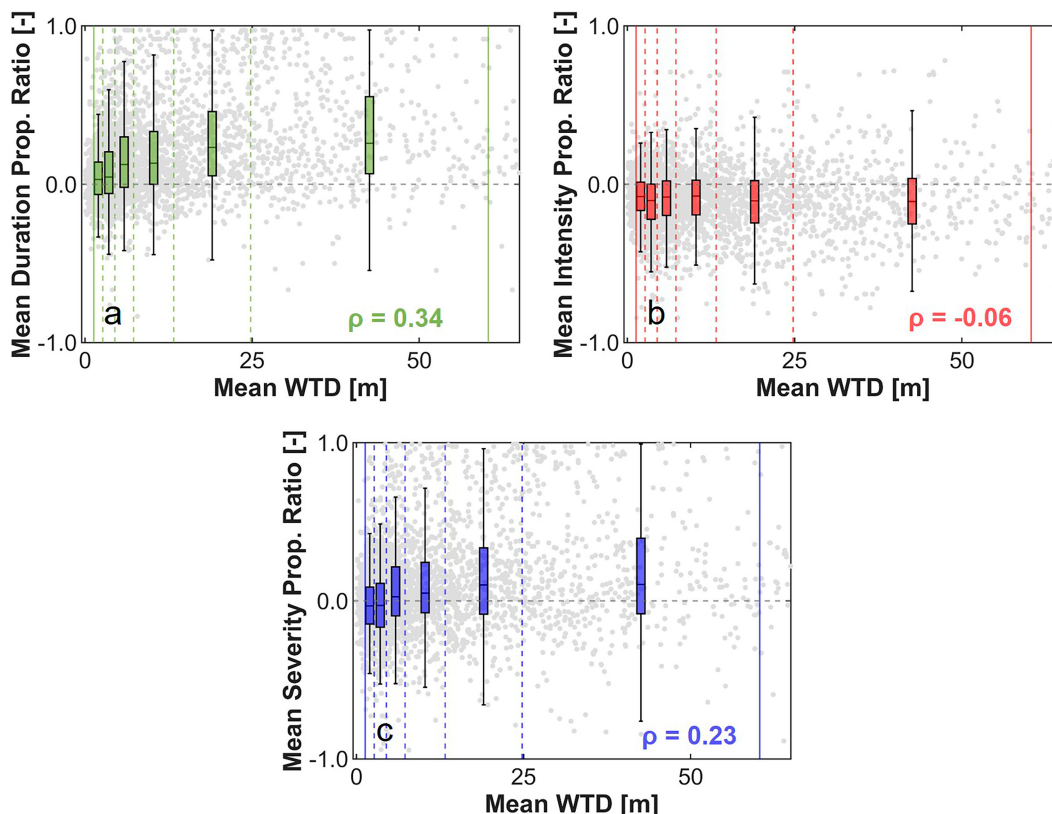


FIG. 10. Relationships between mean WTD and (a) the mean duration propagation ratio, (b) the mean intensity propagation ratio, and (c) the mean severity propagation ratio. The data have again been binned between the 5th and 95th percentiles. A small number of sites are not within the limits of these plots. Spearman rank correlation coefficients ρ are indicated in each case; these have been calculated using all data points and, in all cases, were significant at the 5% significance level.

3) LINKS WITH MEAN WATER-TABLE DEPTH

Deeper water tables were associated with greater increases in both drought durations and severities during propagation. Given the strong correlation between the mean duration and severity propagation ratios ($\rho = 0.81$, supplemental material), it is unsurprising that both showed similar relationships. In contrast, meteorological and groundwater drought durations and severities both tended to be more similar in regions where the water table was shallow. There was no clear relationship between the mean WTD and the mean intensity propagation ratio. This suggests that, while the depth of the water table influenced the degree of drought lengthening, this did not clearly affect the extent to which droughts were attenuated.

The relationships between the mean WTD and the changes in drought duration and severity can be directly linked to the concept of the water-table ratio (WTR), which is a metric that indicates the extent to which the position of the water table is controlled by topography or recharge (Cuthbert et al. 2019; Haitjema and Mitchell-Braker 2005). In regions where the water table is primarily controlled by topography ($WTR > 1$), which typically corresponds to where the water table is shallow (<10 m), interactions between groundwater and the climate

tend to be bidirectional. In these regions, groundwater can both receive recharge from the climate, as well as return moisture back to the atmosphere via evapotranspiration if the water table is sufficiently shallow (Cuthbert et al. 2019). This strong linkage between climate and groundwater may explain the finding that meteorological and groundwater drought durations and severities tended to be similar at sites with shallow mean WTDs.

Conversely, groundwater–climate interactions tend to be mostly unidirectional where the water table is predominantly recharge-controlled ($WTR < 1$). Water tables in these areas are typically deeper, with groundwater being much more disconnected from the land surface. This means that, while groundwater in these areas can receive recharge from the climate, it does not return much back to the atmosphere (Cuthbert et al. 2019). Recharge of deeper groundwater tends to be subject to greater damping. This is particularly the case for high-frequency climatic variability, which may be damped to a greater extent than lower-frequency signals (Moeck et al. 2024; Rust et al. 2019; Tsy-pin et al. 2025; Velasco et al. 2017). This reduced response of deeper groundwater to short-term climatic variability, and therefore the more predominant role of longer-term variations in influencing groundwater levels (Kumar et al. 2016; Li and Rodell 2015; Tsy-pin et al. 2025), may explain the

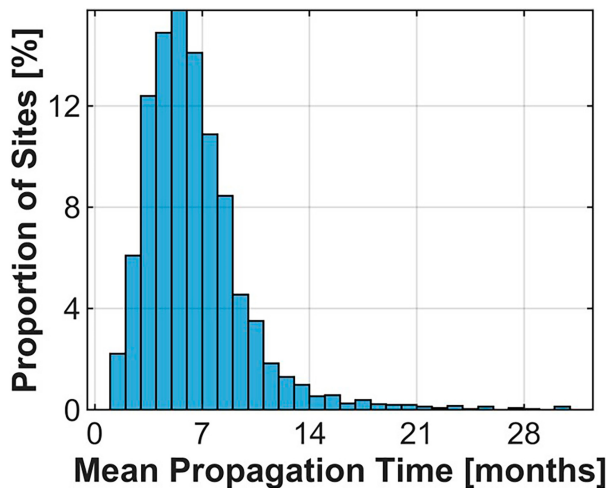


FIG. 11. The distribution of the delay between the onset of meteorological droughts and the corresponding groundwater droughts across the sites.

greater increases in drought durations and severities at these sites. This is because the groundwater drought conditions may not only be due to the single paired meteorological drought but also be a response to longer-term water deficits, reflecting a possible limitation of the drought pairing approach applied in this analysis.

Additionally, it was found that both the mean duration and mean severity propagation ratios were much more variable at higher mean WTDs. This implies that groundwater responses to meteorological drought were much more inconsistent in these areas. For deeper groundwater, this greater variability may be due to differences in hydrogeology. For example, dense fracture networks can enhance connectivity between deeper groundwater and surface conditions (Mihret and Wuletaw 2025). Consequently, meteorological and groundwater drought durations and severities would be expected to be more similar in these areas. A contrasting example would be where there are extensive confined conditions within the aquifer, which would increase the disconnection between groundwater and climate (Wendt et al. 2020), thus leading to decreased groundwater sensitivity to meteorological extremes. Therefore, while deeper groundwater would generally be expected to have a more damped response to climatic variability, differences in local hydrogeology may have led to stronger or weaker connections between the two, resulting in this greater variability.

c. Limitations and outlook for future analysis

This study has provided valuable new insights into drought propagation at continental scales, indicating areas in each continent where groundwater more effectively buffers meteorological droughts. This has important implications for water resource management in these areas. Given the projected increases in drought frequency and intensity in the coming years, such information may become increasingly invaluable.

However, it is important to acknowledge that there were some limitations to this analysis. For example, despite IGRAC's

dataset representing a significant advancement in groundwater data sharing, large areas remain with poor coverage. This is either through a lack of monitoring sites or, alternatively, a lack of sites with data of sufficient quality. Although coverage across North America was greatly improved compared to previous studies (e.g., Li and Rodell 2015; Schreiner-McGraw and Ajami 2021), central and western areas still had relatively poorer coverage compared to the eastern coast. Similarly, coverage was lacking across parts of Europe, including Italy, Poland, and Germany. This limited representation of these climatic and hydrogeologic settings in the analysis. Future studies will benefit from increased data collection and sharing in these regions.

Similarly, this dataset had limitations in the length of the available time series. Hence, SPI and SGI calculations were based on only 20 years of data, which is below the WMO recommendation of 30 years (Kumar et al. 2016; WMO 2012). The SPI and SGI are probability-based drought indices. Calculation using longer time series is advantageous, providing greater confidence in the calculated statistics (Wu et al. 2005). Future advancements in groundwater monitoring and increased data sharing will enable access to longer time series for future analyses, therefore enhancing the robustness of the calculated drought indices. However, despite these limitations, the key features of drought propagation could be observed at these sites in this study. This supports the suitability of using this dataset for future studies of groundwater drought.

Additionally, there are some uncertainties in the propagation ratio approach. This assumes that groundwater droughts are triggered by only the latest meteorological drought. This may not always be valid, for example, where groundwater is significantly affected by human activities, is confined, or the water table is very deep. The impact of including such sites in the analysis was tested using proxy measures due to data limitations (supplemental material). Despite the potentially lesser suitability of the propagation ratio approach for these sites, their inclusion was found to not alter the main conclusions drawn significantly, indicating that this limitation did not affect the analysis presented. However, their effects may be more substantial at smaller spatial scales. Although the propagation ratios were found to be linked to both the mean meteorological drought characteristics and the mean WTD, clear relationships were not found with other variables, including the geology, topography, or climate characteristics. This may suggest that these variables minimally influenced drought propagation. However, it may instead be that their effects were obscured by anthropogenic activities (AghaKouchak et al. 2021). Significant quantities of groundwater are abstracted for human water use globally (Famiglietti 2014). During periods of drought, abstraction can further increase to counteract shortages in surface water availability (Van Loon et al. 2016). Because this modifies the natural response of groundwater to meteorological drought, causing greater declines in groundwater levels than would be expected under natural conditions, this may explain this lack of clear relationships with these other variables. However, this study was limited in its characterization of human impacts due to a lack of abstraction data. Given that human activities are continuing to modify groundwater regimes globally, and that droughts are projected to become more frequent in many regions (IPCC 2023a,b),

there is therefore an urgent need to better understand these anthropogenic effects.

5. Conclusions

In this study, data from more than 3200 groundwater wells across North America and Europe were used to assess groundwater drought. Using observations of water-table depth (WTD) and precipitation data from ERA5-Land, two standardized drought indices (SPI and SGI) were calculated for each site. Based on a pairing approach to match groundwater droughts to preceding meteorological droughts, propagation ratios were calculated to express changes in drought durations, intensities, and severities during drought propagation. Relationships between these propagation ratios and a range of variables were analyzed to identify possible influencing factors. This included the geology, topography, climate characteristics, and mean WTD. Additionally, spatial variability in the resulting drought characteristics was assessed.

There were several key findings. First, groundwater drought characteristics showed much greater spatial variability than that observed in the meteorological drought characteristics. This emphasizes the site-specific nature of the response of groundwater to meteorological drought. Second, droughts tended to lengthen but reduce in intensity during drought propagation. This is consistent with the conceptual understanding of drought propagation, with groundwater showing a smoothed response to meteorological extremes. Third, greater drought lengthening was typically observed at sites with deeper water tables. This was likely due to increased damping of short-term climatic variability and therefore the greater influence of longer-term water deficits on groundwater levels at these sites. Fourth, there tended to be greater drought attenuation at sites where meteorological droughts were more intense. This may be linked to nonlinear processes such as baseflow reduction which constrain groundwater declines during periods of meteorological drought, restricting further groundwater drought intensification. Clear relationships were not found between the propagation ratios and the remaining variables.

Compared to previous studies, which have typically investigated drought propagation using in situ data over smaller regions, this analysis has provided valuable insights into groundwater drought across continental scales. In particular, this analysis has indicated the regions in each continent where groundwater can more effectively buffer periods of meteorological drought. Thus, this provides insights into groundwater vulnerability, with important implications for water resource management. Such information may become particularly important given the projected increases in drought frequency and intensity in the coming years due to climate change. However, there are some limitations to note, for example, poorer coverage of groundwater monitoring sites in certain regions, as well as restrictions in the lengths of groundwater time series. Nevertheless, despite these limitations, the findings of this study are consistent with the conceptual understanding, supporting the suitability of using this dataset for future analyses of groundwater drought.

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Data availability statement. The groundwater data are freely available through IGRAC's GGMN portal (<https://ggis.un-igrac.org/view/ggmn/>, accessed: 30 January 2025). Topographical data were obtained from version 1.03 of MERIT DEM, available through the Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/MERIT_DEM_v1_0_3, accessed: 24 April 2025). ERA5-Land precipitation data are available through the Climate Data Store (<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land-monthly-means>, accessed: 1 November 2024). Potential evapotranspiration data were obtained from version 4.1 of the Global Land Evaporation Amsterdam Model (GLEAM; <https://www.gleam.eu/>, accessed: 24 April 2025). Hydraulic conductivity data were obtained from version 2.0 of the Global Hydrogeology Maps (GLHYMPS, <https://borealisdata.ca/dataset.xhtml?persistentId=doi:10.5683/SP2/TJJIU>, accessed: 16 May 2025). Estimates of population densities and urban and cropland areas were obtained from version 3.3 of the History Database of the Global Environment (HYDE, https://geo.public.data.uu.nl/vault-hyde-data/HYDE%203.3%5B1696419673%5D/original/Baseline_scenario/anthromes/, accessed: 31 March 2025). Estimated water withdrawals across the globe were taken from Huang et al. (2018, <https://zenodo.org/records/1209296>, accessed: 20 March 2026).

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